Harsh Solanki

Gen AI Engineer

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Summary

Generative AI Engineer with proven experience building local-first, privacy-first AI systems from fine-tuning LLMs and deploying RAG pipelines to orchestrating multi-agent workflows on self-hosted GPU infrastructure. Passionate about applying AI to real-world problems. Curious, experimental, and constantly building.

Skills

Programming Languages: Python, Swift, C, HTML
AI/ML Frameworks: PyTorch, TensorFlow, LangGraph, LangChain, Unsloth Studio, Scikit-learn, NumPy, Pandas
Generative AI & LLM: LLM Fine-tuning (QLORA, GRPO), RAG, Prompt Engineering, Hugging Face, Transformers, Diffusion Models, Local LLM Deployment, NLP
Agentic AI & Automation: OpenClaw, MCP, Pydantic-AI, n8n, Google Antigravity, Claude Code, Lovable, Ollama
Vector & Database: Pinecone, FAISS, Chroma, SQLite-vec, PostgreSQL, Supabase
Deployment & Cloud: FastAPI, AWS, GCP, Microsoft Azure, Hugging Face Spaces, Docker, Git, Linux, HomeLab
Soft Skills: Analytical Thinking, Problem-Solving, Team Collaboration, Effective Communication

Experience

Generative AI Intern **Jul 2025 – Sep 2025**
Manav Rachna Centre for Health Innovations

- Architected and managed end-to-end local GPU-accelerated GenAI workflows using Wan2.1, FLUX, and LTX-2, improving content iteration speed by 20% while ensuring full data privacy.
- Built 2 automation pipelines for short-form health awareness video production, delivering 4 videos on stroke, seizure, and other emergencies bridging Generative AI with healthcare communication.
- Produced a product marketing video for the AnemoScan mobile app; collaborated cross-functionally with healthcare and design teams to align AI output with campaign goals.

IT Intern **Jul 2024 – Aug 2024**
NTPC Ltd.

- Gained hands-on exposure to enterprise-scale IT infrastructure, supporting 3 systems and resolving 2 tickets for a team of 5 employees within one of India's largest public sector organizations.
- Created 4 Standard Operating Procedures (SOPs) for recurring hardware issues, improving resolution clarity for internal IT staff.

Projects

Velora: AI-Powered Hi-Fi Music Player | *Gemma 4, Pydantic-AI, FastAPI, SQLite-vec, Swift* **Jul 2026**

- Engineered a local-first iOS music client and custom Python AI engine orchestrated via **FastAPI** and **Pydantic-AI**, enabling sub-second natural language track search without external API cloud costs.
- Fine-tuned a quantized **Gemma 4 12B** model using **Unsloth** and GRPO, reducing VRAM footprint by ∼40% (down to 7GB) while maintaining full logical reasoning capacity for semantic track selection.
- Architected an ultra-lean memory layer using `sqlite-vec` and plain-text Obsidian Markdown files, bypassing massive graph DBs to achieve zero-setup, hyper-fast audio embedding (CLAP) lookups.

PersonaFlow - AI Wellness Companion | *RAG, Supabase, n8n, Ollama, Vercel* **Dec 2025**

- Built a full-stack platform featuring an empathetic AI companion powered by hermes3:8b-llama3.1, a model for emotionally intelligent, supportive dialogue.
- Implemented long-term memory using Supabase to store journals, habits, and reflections enabling the AI to develop a persistent "second self" that understands the user more deeply over time.
- Built advanced AI pipelines using RAG and multi-tool agents to deliver context-aware, emotionally aligned guidance that becomes more accurate and personalized with every interaction.

– Engineered a privacy-first, 100% local AI system ensuring complete data ownership while supporting real-time semantic search and multi-step reasoning workflows through n8n automation.

NutriBud AI Nutrition Assistant | *LLAMA 3.2 Vision, Unsloth, n8n* **Sep 2025**

– Built an end-to-end Telegram Bot for automated meal photo analysis to estimate calories, protein, fat, and carbs entirely using self-hosted n8n and Telegram Bot API.
– Significantly enhanced model accuracy by 50% for diverse cuisine by fine-tuning LLAMA 3.2 Vision 11B on the Indian Food Images Dataset, leveraging Unsloth/QLORA for memory-optimized training on 12 GB VRAM.
– Open-sourced the fine-tuned model on Hugging Face that substantially outperformed general VLMs on region-specific food classification accuracy.

Self-Hosted AI Infrastructure Rig | *HiveOS, Salad, Ubuntu, Vast.ai* **Dec 2024**

– Built and managed a 4x NVIDIA RTX GPU HomeLab rig with 24/7 uptime for self-hosting AI/ML and compute monetization via Vast.ai.
– Optimized thermal management and power configuration to reduce energy consumption by 15% while maintaining peak GPU throughput.

Awards & Certifications

- **Microsoft:** Azure Data Scientist Associate (DP-100) (2025), Azure Data Engineer Associate (DP-203) (2025), Azure AI Fundamentals (AI-900) (2024), Azure Fundamentals (AZ-900) (2024), Microsoft Security, Compliance & Identity (SC-900) (2024)
- **Securiti:** AI Security & Governance Certified
- **Tutedude:** Generative AI Course (Jun 2026)
- **Infosys Springboard:** Machine Learning using Python (2024), Python for Data Science (2024)

Education

B.Tech Computer Science (AI/ML Specialization) **Graduating: Jul 2026**
Manav Rachna International Institute of Research and Studies *CGPA: 7.2 / 10*